Sahil Sahran

sahilsahrans@gmail.com • (403)-923-1710

EDUCATION

Bachelor of Applied Science and Engineering (B.A.Sc) in Electrical Engineering + PEY Co-op *Sep. 2025 – Apr. 2030 (expected)*
University of Toronto, Toronto, ON cGPA: 3.97

Ontario Secondary School Diploma & International Baccalaureate Diploma *Sep. 2022 – Aug. 2025*
Maple High School, Vaughan, ON

EXPERIENCE

D2Woodworking *Markham, ON*
CNC Programmer May *2026 – Present*

- Delivered dozens of custom cabinet and shelving builds on time and within budget by programming precise cut paths in Mozaik CAD/CAM software and translating blueprints into optimized G-code toolpaths
- Integrated soft-close actuators and under-cabinet LED systems into custom cabinet builds while maintaining compatibility with existing electrical infrastructure
- Minimized offcut waste on each project as measured by material yield per sheet by optimizing nesting layouts and cut sequencing for hardwood and plywood panels

Canada's Wonderland *Vaughan, ON*
Games Associate *Apr. 2022 – Oct. 2025*

- Ranked #1 in upsell revenue among all park game sections as measured by end-of-season sales totals by applying real-time guest engagement and persuasive communication techniques
- Organized coworkers by rotating positions and covering stations to ensure smooth operation, maintaining the team's efficiency at all times
- Delivered excellent customer service to earn the G.O.A.T. Service recognition, improving overall guest satisfaction

Top Radiation Safety Consultancy *Vaughan, ON*
Admin Assistant *Nov. 2024 – Present*

- Conducted leak testing of nuclear gauge samples to detect potential radiation hazards
- Maintained zero compliance violations across all audited client files as measured by regulatory review outcomes by systematically evaluating measurement logs against federal radiation protection standards
- Prepared compliance certificates, providing assurance for clients and customers

PROJECTS

Developing an ESP-32 Based Sound/Proximity Recognition Assistant (ESPRA) *May 2026 –Jun. 2026*

- Accomplished a unified control architecture and voice pipeline, achieving an end-to-end response under 2 seconds with zero state crashes, by developing dual-core FreeRTOS event queues, on-device wake-word inference linked to remote LLM server, and static PSRAM buffers on a single ESP32-S3
- Eliminated switching noise, EMI, and hardware boot faults, resulting in clean peripheral signal integrity and zero servo twitches, by designing a three-rail isolated buck-converter power network, a star-ground topology, and a P-channel MOSFET power gate
- Implemented closed-loop locomotion and power protection, maintaining a straight-line drift under 5cm/m and automated undervoltage shutdowns, by programming a 50Hz PI encoder loop, interfacing a ToF sensor over hardware I2C, and reading calibrated analog voltage dividers

Developing Beehive Wintering Monitoring System *Jan. 2026 – Apr. 2026*

- Enabled full non-invasive environmental monitoring as measured by sensor coverage of all target parameters by designing an ESP32-based embedded system integrating temperature, humidity, and gas sensors
- Analyzed electromagnetic field exposure constraints in biologically sensitive systems, deriving a theoretical safe distance by comparing the isotropic power propagation against biological thresholds
- Developed and executed controlled experiments (thermal, humidity, electromagnetic field) and quantified response to transient changes to verify sensor reliability and system safety

Report on Modifying JPEG Compression Algorithm to Allow for Desired File Size *Aug. 2024 – Nov. 2024*

- Modified the JPEG compression algorithm by implementing Lagrangian Optimization, cosine transform, and quantization in MATLAB, achieving target file size with minimized quality loss
- Produced a report through heavy research and mentor consultation, resulting in an evidence supported project that met academic standards

SKILLS

Technical: MATLAB, C, Python, radiation contamination monitoring, SketchUp, Mozaik CAD/CAM, AutoCAD Fusion, embedded systems and firmware architecture, network pipelines
Professional: Technical writing, teamwork, customer service, mentoring
Languages: English (native proficiency), French (working proficiency)